

Olist Brazil E-Commerce Analysis

SQL + Power BI Case Study

1. Project Overview

Introduction

E-commerce platforms generate large amounts of transactional data every day. Analyzing this data can help businesses understand customer behavior, identify growth opportunities and improve operational performance.

In this project, I analyzed the Olist Brazil E-Commerce dataset using SQL and Power BI to uncover patterns in customer purchasing behavior, revenue generation, logistics performance and payment preferences.

The dataset contains more than 100,000 orders placed between September 2016 and October 2018 across multiple Brazilian states and cities. Through a combination of SQL-based exploration and interactive Power BI dashboards, the objective was to transform raw transactional data into meaningful business insights.

Project Objectives

The analysis focuses on answering the following business questions:

- How did order volume and revenue evolve over time?
- Which regions contributed the most customers and revenue?
- What role did freight costs play in overall performance?
- How efficient was the delivery process across different states?
- How do customers prefer to pay for their purchases?
- What business opportunities can be identified from the data?

Tools Used

- SQL: - Data Analysis, aggregations and business analysis
 - Power BI: - Dashboard design and business reporting
 - DAX: - KPI creation and metric calculations
 - GitHub: - Project documentation and portfolio presentation
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2. Dataset Overview

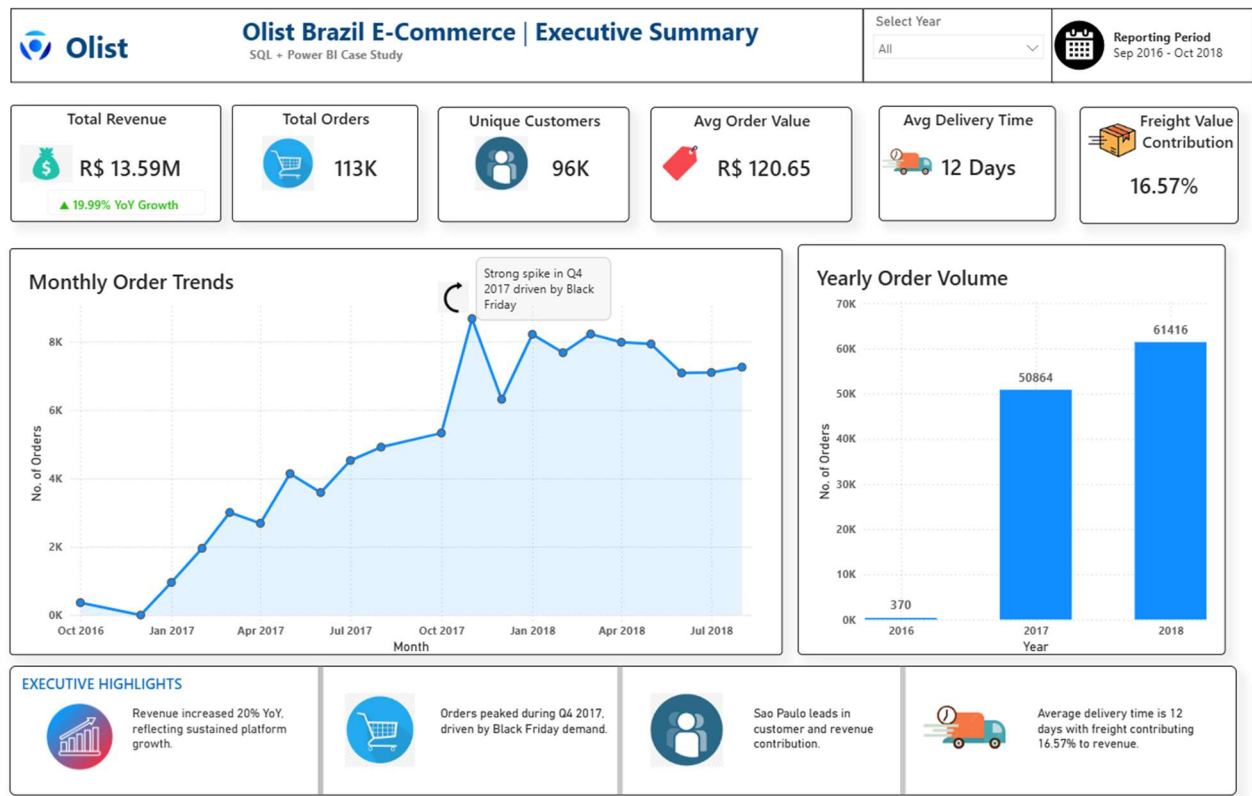
Dataset Coverage

Metric	Value
Reporting Period	Sep 2016 – Oct 2018
Total Orders	113K
Unique Customers	96K
Cities Covered	4,119
States Covered	27
Total Revenue	R\$ 13.59M

Key Business Areas Analyzed

1. Order Performance
 2. Customer & Regional Analysis
 3. Revenue & Freight Analysis
 4. Delivery & Logistics Analysis
 5. Payment Behavior Analysis
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3. Executive Dashboard Summary



The Executive Dashboard provides a consolidated business overview of Olist's performance.

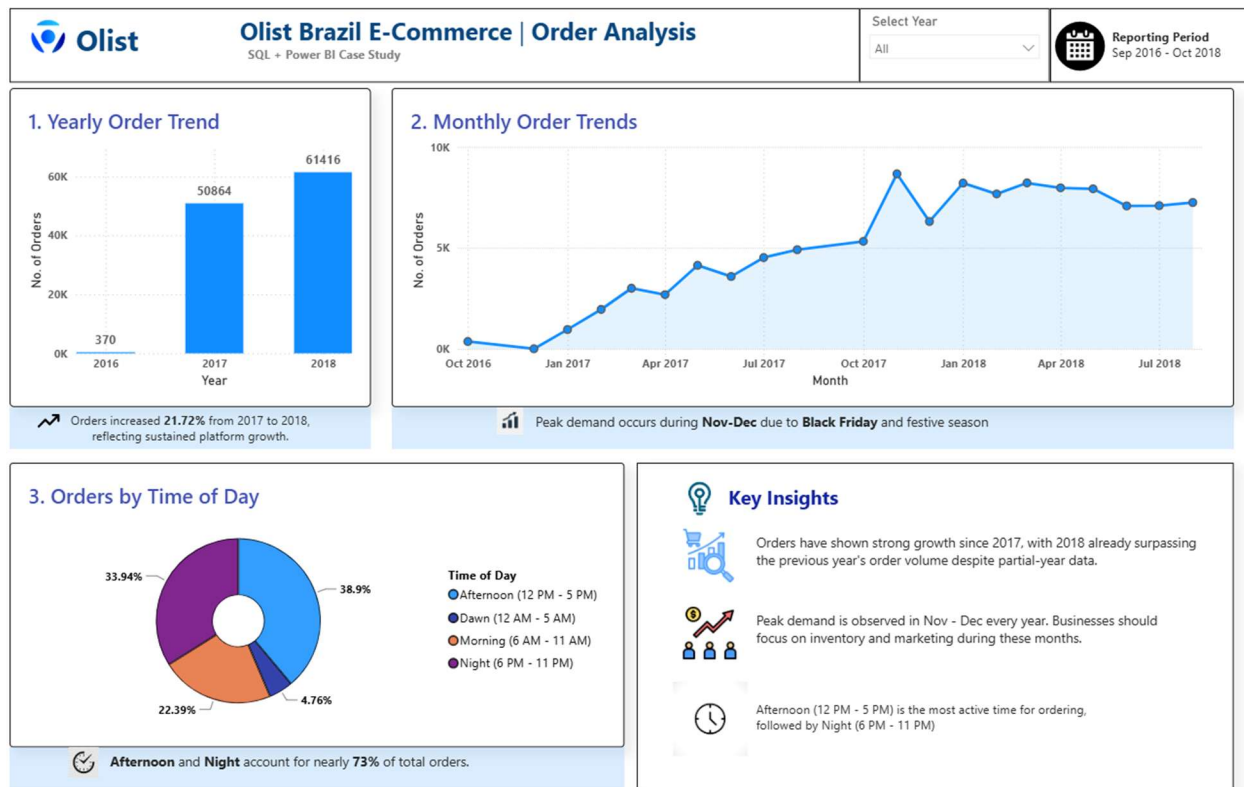
Key Findings

- Revenue exceeded R\$13.5M during the reporting period.
- Order volume increased significantly between 2017 and 2018.
- São Paulo emerged as the dominant market in both customers and revenue.
- Average delivery time remained at approximately 12 days.
- Freight represented 16.57% of total product value.

Key Insight

The platform experienced rapid growth while maintaining healthy delivery performance and strong customer acquisition across Brazil.

4. Order Analysis



Key Findings

- Orders increased substantially after 2016.
- Strong seasonal demand was observed during Q4 2017.
- Black Friday and festive periods generated the highest order activity.
- Afternoon and Night represented the most active purchasing periods.

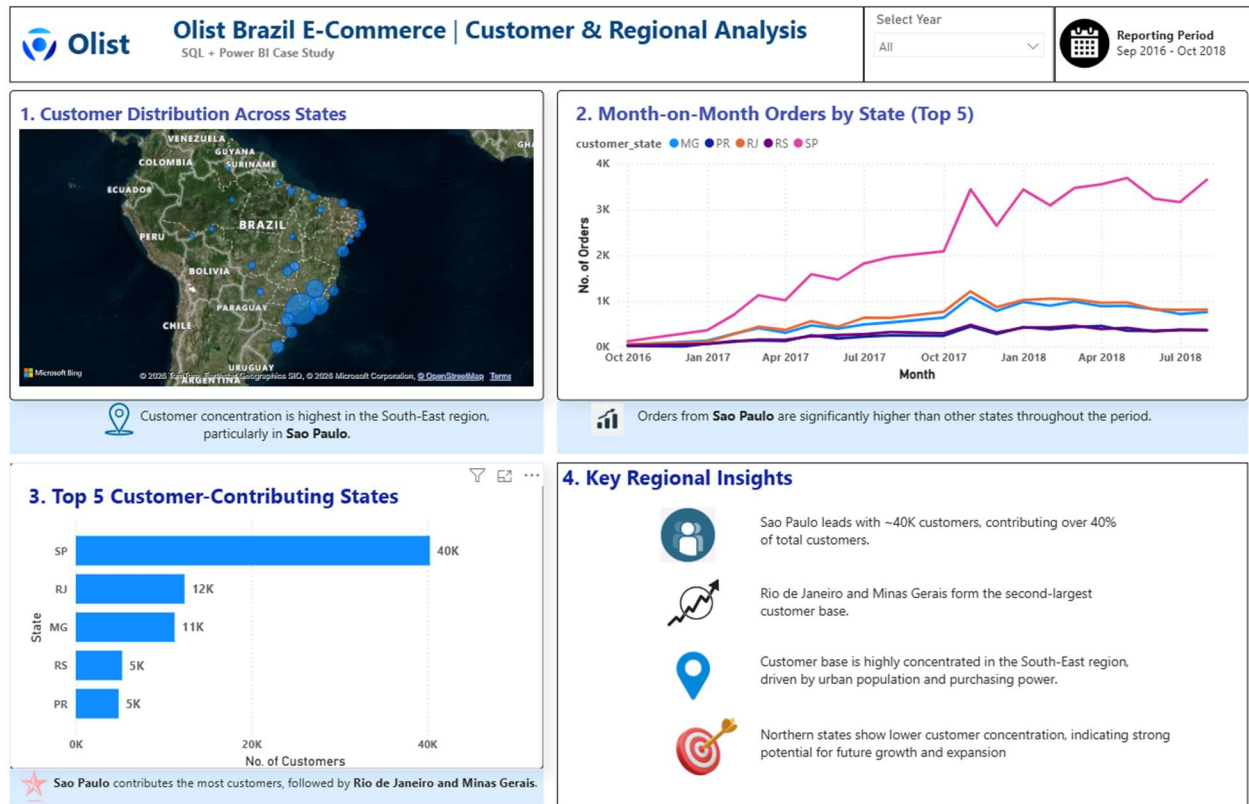
Key Insight

Customer purchasing behavior is highly influenced by seasonal events. Promotional campaigns launched during festive periods can significantly improve sales performance.

Business Recommendation

Focus marketing investments during Q4 events such as Black Friday, Christmas and New Year campaigns.

5. Customer & Regional Analysis



Key Findings

- São Paulo contributed the largest customer base.
- Rio de Janeiro and Minas Gerais formed the next largest markets.
- Customer concentration was heavily skewed toward South-East Brazil.
- Northern states contributed comparatively fewer customers.

Key Insight

Customer acquisition is strongly concentrated within economically developed regions that benefit from better infrastructure, higher internet adoption and stronger purchasing power.

Business Recommendation

Continue strengthening customer retention in major states while gradually expanding into underpenetrated regions.

6. Revenue & Freight Analysis



Key Findings

- Total Revenue Generated: R\$13.59M
- Average Order Value: R\$120.65
- Freight Contribution: 16.57%
- São Paulo generated more than R\$5M in revenue.

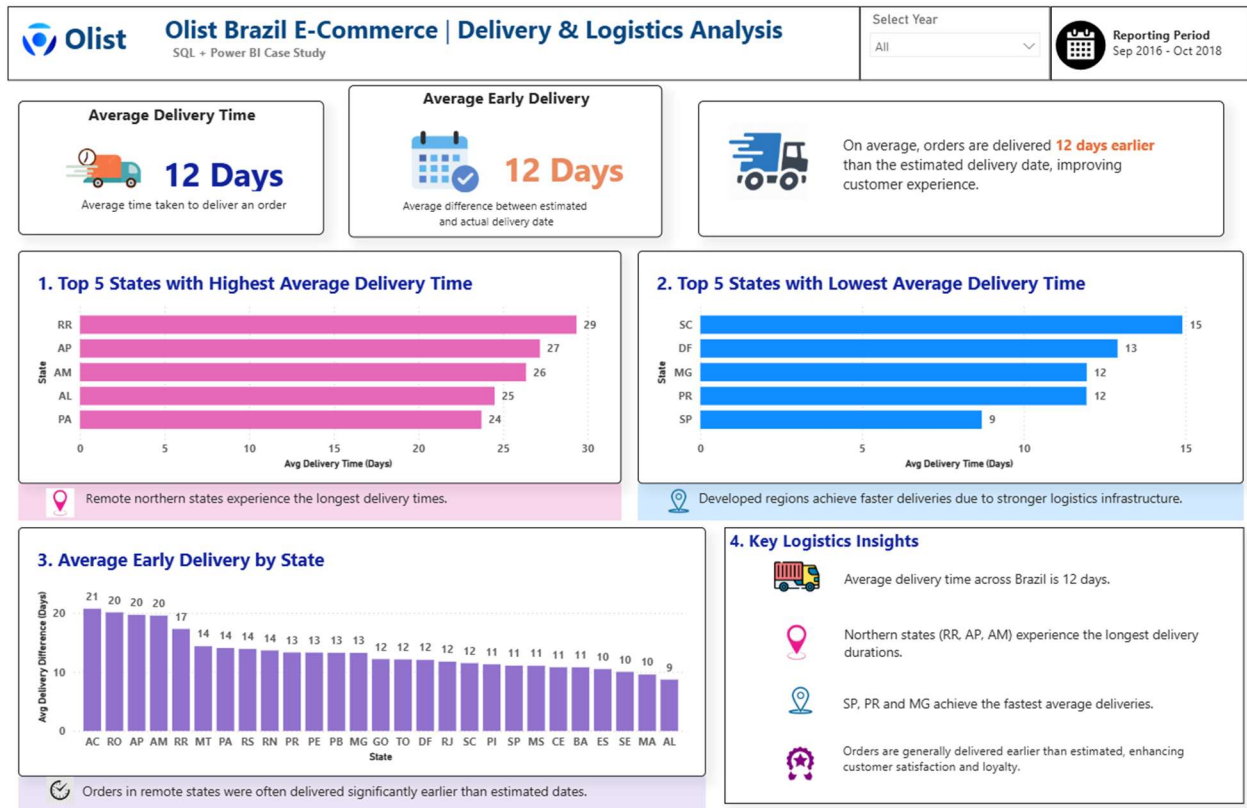
Key Insight

Revenue generation is heavily concentrated in a few key states. States with lower freight costs generally tend to exhibit higher revenue generation and customer activity.

Business Recommendation

Optimize logistics networks and warehouse placement to reduce freight expenses in lower-performing regions.

7. Delivery & Logistics Analysis



Key Findings

- Average Delivery Time: 12 Days
- Orders were delivered approximately 11-12 days earlier than estimated on average.
- Northern states recorded the highest delivery durations.
- South-East states consistently achieved faster deliveries.

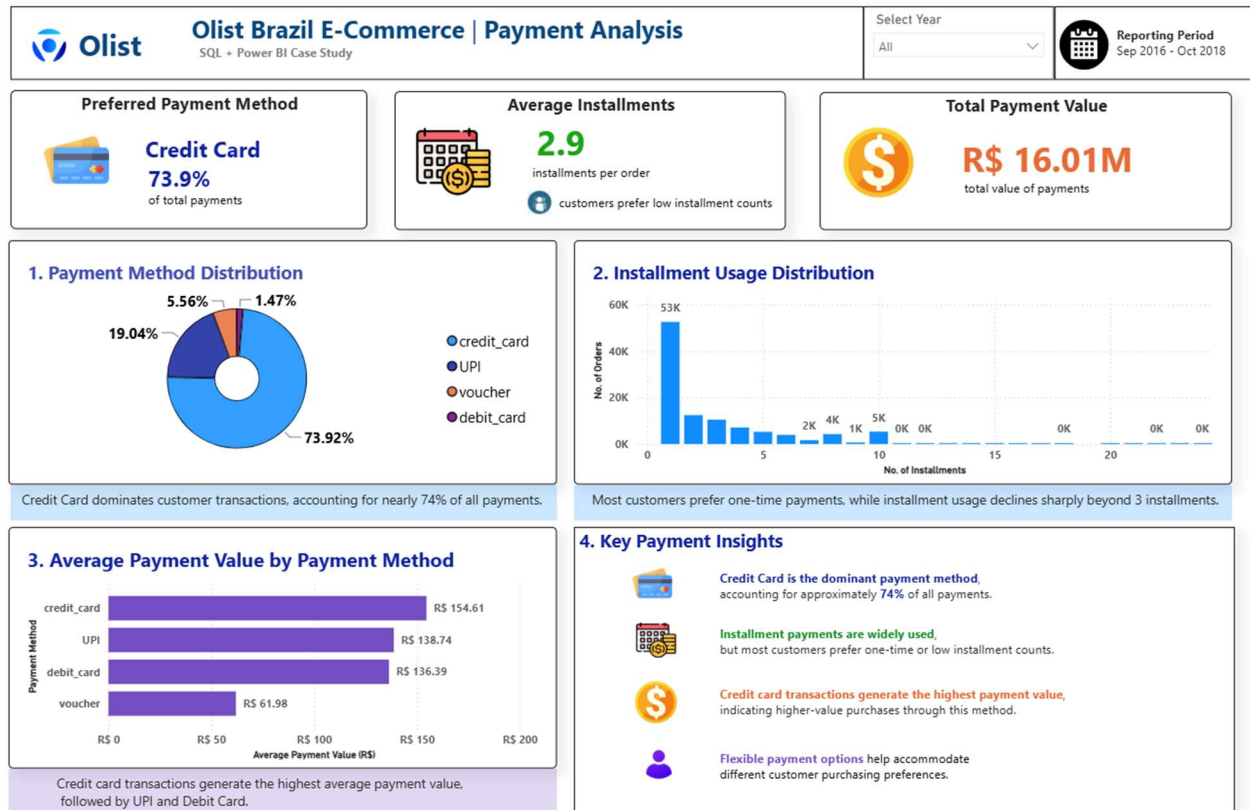
Key Insight

The logistics network performs efficiently despite Brazil's geographic diversity. Conservative delivery estimates appear to contribute positively to customer satisfaction.

Business Recommendation

Focus on improving delivery infrastructure within remote regions to reduce fulfilment times.

8. Payment Behavior Analysis



Key Findings

- Credit Card accounted for approximately 74% of all payments.
- Average Installments per Order: 2.9
- Most customers preferred one-time or low-installment purchases.
- Credit Card transactions generated the highest average payment value.

Key Insight

Customers demonstrate a strong preference for convenient and flexible digital payment options, with credit cards dominating payment activity across the platform.

Business Recommendation

Leverage payment partnerships and targeted offers to encourage higher-value purchases and improve customer conversion rates.

9. Final Business Recommendations

1. Strengthen High-Performing Markets

Prioritize São Paulo, Rio de Janeiro and Minas Gerais through loyalty programs, targeted marketing and customer retention initiatives.

2. Expand Strategically

Develop logistics partnerships and regional campaigns to improve penetration in underrepresented states.

3. Capitalize on Seasonal Demand

Increase promotional activities during Black Friday, Christmas and New Year periods.

4. Improve Logistics Efficiency

Reduce freight costs through warehouse optimization and better regional fulfilment strategies.

5. Enhance Payment Experience

Promote digital payment options and installment plans that align with customer purchasing behavior.

10. Conclusion

This project provided a comprehensive view of Olist's e-commerce operations between 2016 and 2018.

The analysis revealed strong growth in both orders and revenue, with São Paulo consistently emerging as the platform's most important market. Customer activity was heavily concentrated within South-East Brazil, while several regions presented opportunities for future expansion.

From an operational perspective, the logistics network performed efficiently, with most orders being delivered earlier than the estimated delivery date. However, delivery times remained noticeably longer in remote northern states, highlighting potential areas for logistical improvement.

Payment analysis showed a clear customer preference for credit card transactions and low-installment purchases, suggesting that convenience and payment flexibility play an important role in purchasing behavior.

Overall, the project demonstrates how data can be used to understand customer behavior, evaluate operational performance and support business decision-making. By combining SQL analysis with interactive Power BI dashboards, the findings were translated into actionable insights that could help improve customer experience, optimize logistics and support future growth initiatives.